

Motion Detection

USING COMPUTER VISION

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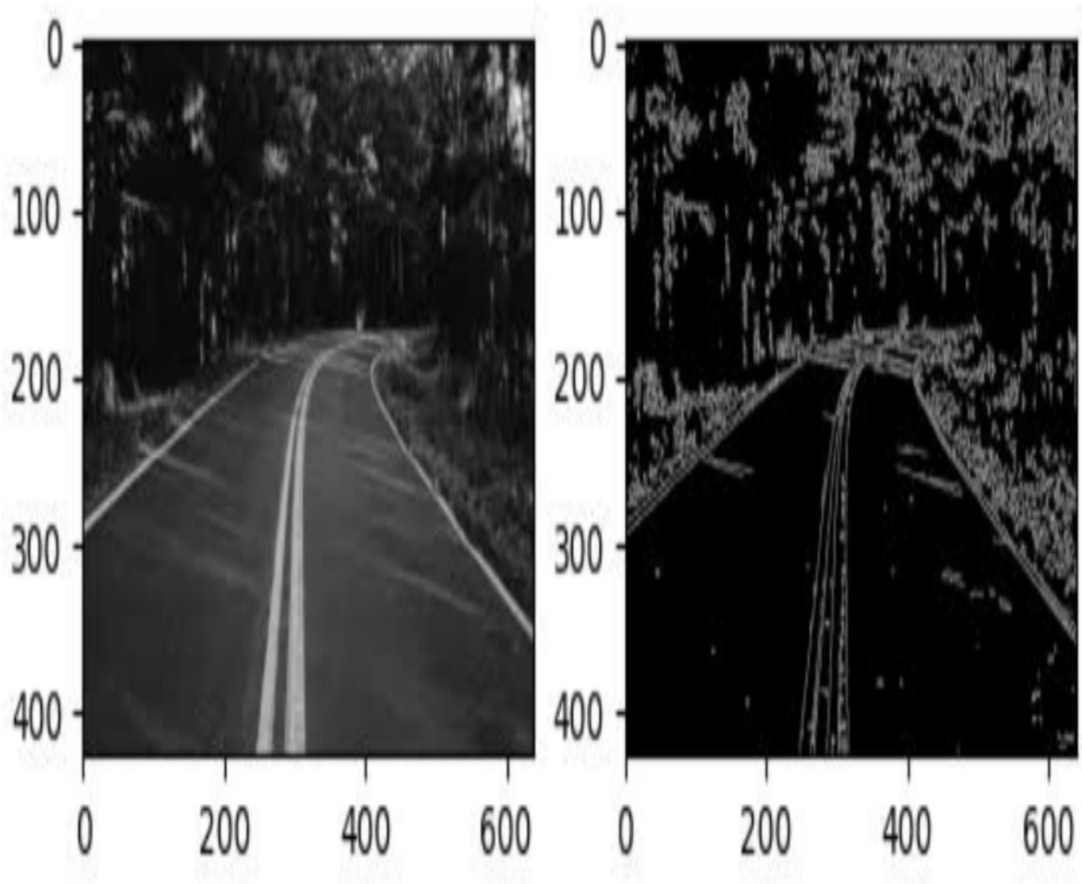
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INTRODUCTION



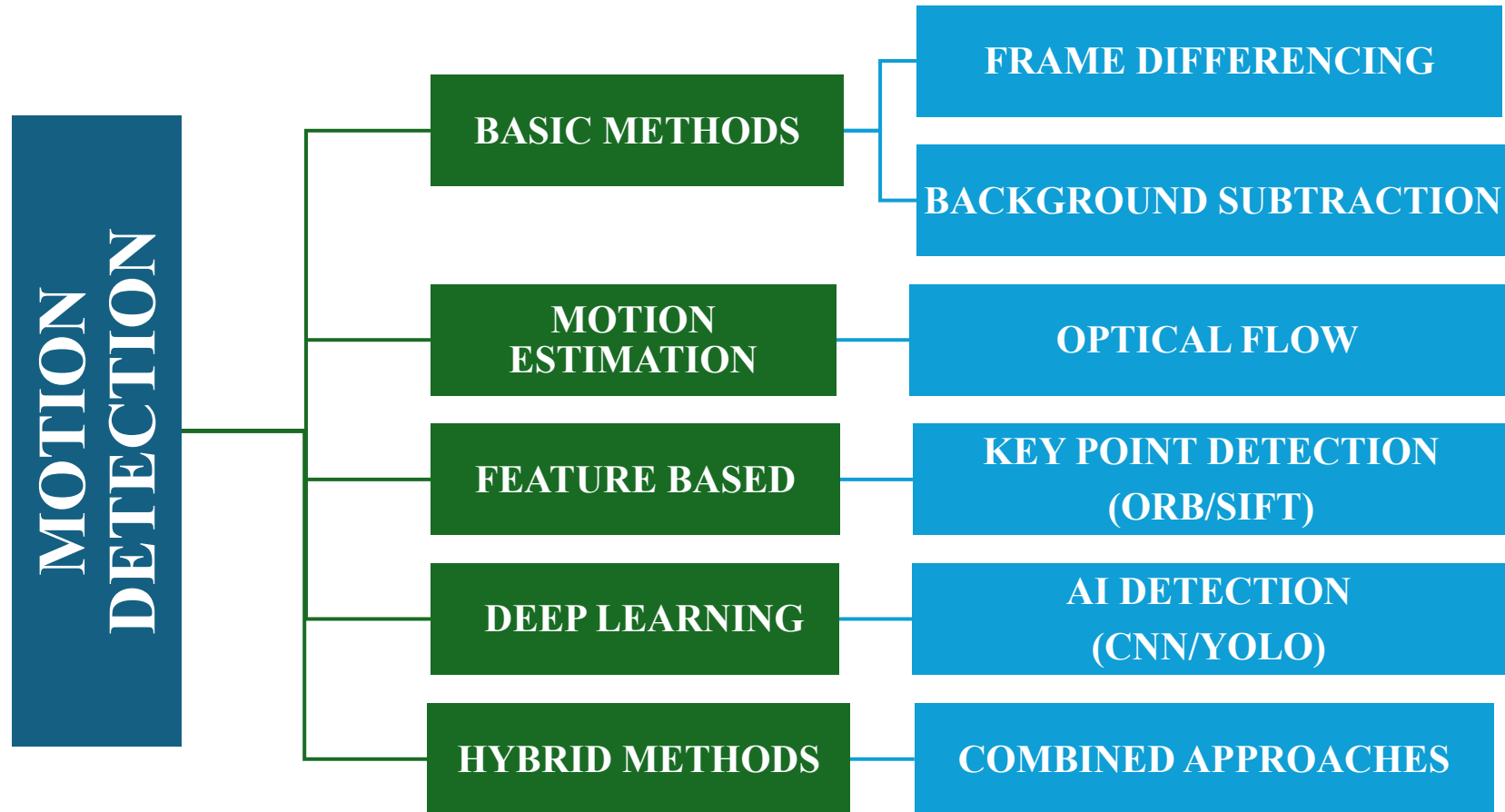
- **Motion detection is a computer vision technique used to detect movement in video streams.**
- **The method helps in identifying moving objects while ignoring static background.**
- **It enables automatic, real-time monitoring with minimal human effort.**
- **It operates by analysing variations between consecutive video frames to distinguish moving objects from a static background.**
- **It forms the basis for higher-level tasks such as object tracking, behavior analysis, and activity recognition.**
- **Provides visual feedback using thresholding and audio alert (beep) with user-controlled toggle.**
- **It is widely used in surveillance, traffic monitoring, and smart security systems.**

MOTIVATION



- ❖ Growing need for better security and surveillance
- ❖ Manual monitoring is time-consuming and inefficient
- ❖ Need for real-time detection of suspicious activities
- ❖ Helps in reducing human effort and errors
- ❖ Useful in smart homes and smart cities
- ❖ Provides real time alert(beep) for detected motion

EXISTING METHODS



CHALLENGES



FIG: DYNAMIC ENVIRONMENTS

- ❑ Hard to separate moving objects from changing background
- ❑ Problems: shadows, lighting changes, noise



FIG:CONTINUOUS BEEP ALERTS

- ❑ Continuous beep alerts can be distracting
- ❑ User control is needed to stop the alert



FIG: BINARY THRESHOLDING

- ❑ Small pixel changes (noise, lighting flicker) can trigger false alarms
- ❑ Fixed threshold causes false detections



FIG: SENSOR VARIABILITY

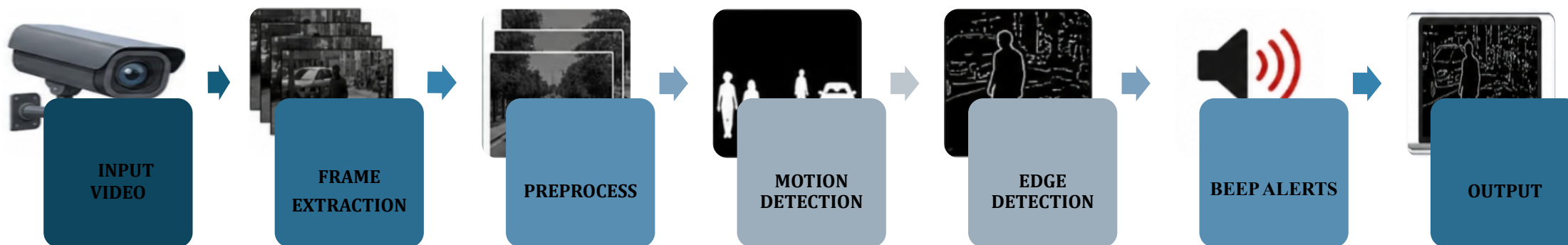
- ❑ Day/night and weather affect detection
- ❑ Camera noise and blur reduce accuracy



FIG:OCCULSION AND TRACEABILITY

- ❑ Objects get hidden by others
- ❑ Difficult to track continuously

PROPOSED APPROACH



SIMPLE IMPLEMENTATION



```
if alarm_mode:
    frame_bw = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
    frame_bw = cv2.GaussianBlur(frame_bw, (5, 5), 0)

    difference = cv2.absdiff(frame_bw, start_frame)
    threshold = cv2.threshold(difference, 25, 255, cv2.THRESH_BINARY)[1]
    start_frame = frame_bw

    if threshold.sum() > 300:
        alarm_counter += 1
    else:
        if alarm_counter > 0:
            alarm_counter -= 1

    cv2.imshow("Cam", threshold)
else:
    cv2.imshow("Cam", frame)
```

- ✓ Frame differencing is used to detect motion
- ✓ Frames are converted to grayscale
- ✓ Gaussian blur removes noise
- ✓ Thresholding highlights moving region
- ✓ Motion is detected based on pixel intensity changes
- ✓ Beep alert is triggered when motion exceeds threshold

ALGORITHM

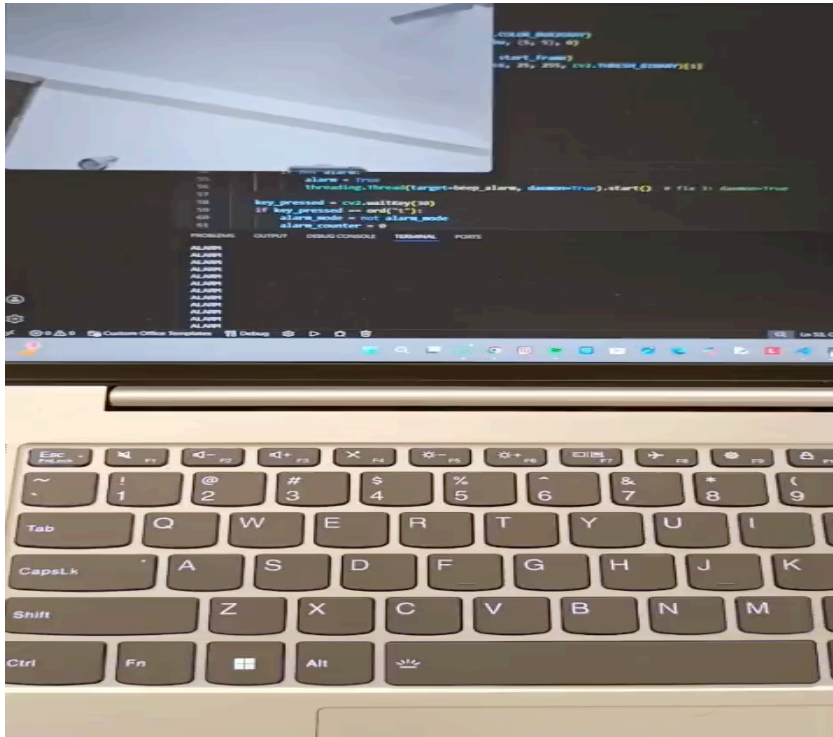


1. Start
2. Capture video from camera
3. Display normal camera feed
4. Wait for user input ('T' key)
5. Press 'T' to toggle motion detection ON/OFF
6. Convert frames to grayscale
7. Apply Gaussian Blur
8. Compute difference between frames
9. Apply thresholding (black & white output)
10. Update reference frame
11. Check motion using pixel intensity
12. If motion detected → Increase counter
13. If counter exceeds limit → Trigger beep alert
14. Display output frame (black & white)
15. Repeat steps 2–14 for each new frame
16. If 'Q' is pressed → Stop and release camera

RESULTS



VID: MOTION DETECTION WITH BEEP ALERT



- ✓ Real-time motion detection achieved
- ✓ Accurate detection using frame differencing
- ✓ Black and white output highlights movement
- ✓ Beep alert triggered on motion detection
- ✓ Works effectively in simple environments
- ✓ Real-time processing achieved using OpenCV

APPLICATIONS



**1. SURVEILLANCE/
SECURITY**



2. ROOM MONITORING SYSTEMS



3. SMART HOMES



4. OFFICE/STORE MONITORING



5. NIGHT /LOWLIGHT MONITORING



6. ALARM SYSTEMS



CONCLUSION

- ✓ A basic motion detection system was implemented using frame differencing.
- ✓ The system accurately detects motion and triggers an alert (beep) when activity is detected.
- ✓ Simple techniques such as grayscale conversion, Gaussian blur, and thresholding were used effectively.
- ✓ The system is efficient, lightweight, and suitable for basic surveillance applications.
- ✓ This project demonstrates a simple yet practical approach to real-time motion detection.

REFERENCES



- **IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, Vancouver, Canada.**
- **“Real-Time Motion Detection Using Frame Differencing,” IEEE, 2020.**
- **OpenCV Documentation – <https://docs.opencv.org/4.x/>**
- **Python Documentation – <https://www.python.org/>**
- **GeeksforGeeks : “Motion Detection using OpenCV.” Available at: <https://www.geeksforgeeks.org/python/webcam-motion-detector-python/>**



THANK YOU